

SP19-1: R-11 Atlas Computation — Research Report

Lead: Elie + Lyra Date: May 13, 2026

What R-11 Requires

Verify: for every non-tempered Arthur parameter ψ on $SO(5,2)$, the archimedean A-packet is empty. This means no non-tempered representation contributes to the discrete spectrum.

The intertwining operator sign: $\epsilon_{\text{inf}}(\psi) = (-1)^S$, where $S = \sum_i n_i * \text{floor}((d_i-1)/2)$. Kottwitz sign $\epsilon(SO(5,2)) = -1$ requires S odd. All $d_{\text{max}} \leq 2$ types have $S = 0$ (even) — automatic mismatch.

Current State — What's Already Done

- Toy 2063 (10/13): Enumerates 37 shapes. IW sign kills 23/37. 14 survive needing spectral gap.
- Toy 2067 (13/13): 61 types with self-duality. IW sign kills 39. Unitarity kills 1. $C_2=6$ gap kills 13. Three-layer elimination complete.
- Paper #103 v0.7: Proposition 2.2 complementary filter — Class A ($d_{\text{max}} \leq 2$, killed by Kottwitz sign) + Class B ($d_{\text{max}} \geq 3$, killed by Moeglin). No Atlas needed IF citations verify.
- R-11 Elimination Lemma: Draft with citation chain. [VERIFY] tags remain on exact proposition numbers.

Key Finding: Atlas May Not Be Needed

Paper #103's complementary filter argument: - Class A ($d_{\text{max}} \leq 2$): $S = 0$ always ($\text{floor}((d-1)/2) = 0$ for $d=1,2$). $\epsilon_{\text{inf}} = +1$ mismatches Kottwitz -1 . Pure arithmetic. No Atlas needed. - Class B ($d_{\text{max}} \geq 3$): Moeglin [Moe08] Theorem 1.1 — these contribute only to residual spectrum, not cuspidal. Citation, not computation.

If both hold: NO Atlas computation needed. The argument is purely structural.

Atlas Software Status

NOT installed on this system. Would need source build from liegroups.org (~20 min with Xcode). SageMath, LiE also not available. `sympy.liealgebras` has only basic root system data.

Plan for Elie — Two Options

Option A (Preferred): Pure Python + Citation Pinning

1. Definitive toy: Implement Proposition 2.2 complementary filter completely:
 - Enumerate all 37 shapes (settle the 37/45/61 discrepancy)
 - Partition into Class A ($d_{\text{max}} \leq 2$) and Class B ($d_{\text{max}} \geq 3$)
 - Verify Class A: confirm $S=0$ for every $d_{\text{max}} \leq 2$ type
 - Verify Class B: list all $d_{\text{max}} \geq 3$ types, confirm Moeglin exclusion applies
 - Verify: A union B = all types, no gap
2. Pin citations:
 - Arthur [Art13] Theorem 1.5.1 (global multiplicity formula)
 - Kottwitz sign: $\epsilon(SO(5,2)) = (-1)^{q(G_R)} = -1$. Resolve q formula discrepancy (both give -1).
 - Moeglin [Moe08] Theorem 1.1 for $d_{\text{max}} \geq 3$ exclusion
3. Resolve type count: 37 shapes (Paper #103) vs 61 with self-duality (Toy 2067). Document which count is correct and why.

Option B: Atlas Installation

Install from source, compute A-packets directly. Machine-verifiable answer for every type. Definitive but requires build setup.

Obstacles

1. Kottwitz sign formula inconsistency: Two different formulas, same answer (-1). Pin the correct one.
2. Type count discrepancy: 37 vs 45 vs 61. Paper #103 uses 37.
3. Step 3 dual arguments: Bergman gap ($C_2=6$) vs Mœglin exclusion — complementary but independent. Which is primary?
4. [VERIFY] tag in Paper #103: Exact Arthur proposition number for B_2 root system not yet pinned.

Impact

R-11 verified -> Generalized Ramanujan for $SO(5,2)$ (SP19-5) -> Selberg eigenvalue exact (SP19-6) -> RH/YM/BSD upgrade from conditional to unconditional. One computation, three Millennium upgrades.